Unconstrained Nonlinear Programming Problem

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1. Introduction

An *unconstrained nonlinear programming problem (NLPP)* seeks to find the point $x \in \mathbb{R}^n$ that minimizes (or maximizes) a real-valued function

$$\min_{x\in\mathbb{R}^n} f(x),$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable.

When the objective function $f(x)$ is a **quadratic function**, the problem is called a *quadratic programming problem (QPP)* without constraints.

2. Quadratic Objective Function in Matrix Form

A general quadratic function of n variables can be written as

$$f(x) = x^TQx + c^Tx + d = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}x_ix_j + \sum_{i=1}^{n} c_ix_i + d,$$

where a_{ij} are real constants satisfying $a_{ij} = a_{ji}$ for all i, j. where:

- $x = (x_1, x_2, \ldots, x_n)^T$ is the vector of decision variables,
- $Q = (a_{ij})_{i\times j}$ is an $n \times n$ symmetric matrix of constants,
- $c = (c_1, c_2, \cdots, c_n)^T$ is an n-dimensional column vector, and
- d is a scalar constant.

Problem 1 *Consider the quadratic expression*

$$f(x_1, x_2, x_3) = 3x_1^2 + 2x_2^2 + x_3^2 + 4x_1x_2 - 2x_2x_3.$$

Write the above expression in matrix representation.

Answer: The matrix representation of the given quadratic form $f(x) = x^TQx$. The symmetric matrix Q is given by

$$Q = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}.$$

Hence,

$$f(x) = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 3 & 2 & 0 \\ 2 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

3. Nature of a Quadratic Form

The **nature of a quadratic form** is determined by the signs of its values for all nonzero vectors $x \in \mathbb{R}^n$. Let $Q(x) = x^T Ax$, where A is a symmetric matrix. The form is classified as follows:

(i) Positive Definite:	$Q(x) > 0 \quad \forall\, x \neq 0,$
(ii) Negative Definite:	$Q(x) < 0 \quad \forall\, x \neq 0,$
(iii) Positive Semi-Definite:	$Q(x) \geq 0 \quad \forall\, x,$
(iv) Negative Semi-Definite:	$Q(x) \leq 0 \quad \forall\, x,$
(v) Indefinite:	$Q(x)$ takes both positive and negative values.

Tests for Definiteness

Since A is symmetric, its definiteness can be determined using one of the following equivalent methods:

1. **Principal Minor Test (Sylvester's Criterion):** Let Δ_k denote the kth leading principal minor of A:

$$\Delta_k = \det(A_k), \quad \text{where } A_k \text{ is the leading } k \times k \text{ submatrix of } A.$$

   Then:

$$\begin{cases} A \text{ is positive definite} & \text{if } \Delta_k > 0 \text{ for all } k, \\ A \text{ is positive semi-definite} & \text{if } \Delta_k \geq 0 \text{ for all } k, \\ A \text{ is negative definite} & \text{if } (-1)^k \Delta_k > 0 \text{ for all } k, \\ A \text{ is negative semi-definite} & \text{if } (-1)^k \Delta_k \geq 0 \text{ for all } k, \\ A \text{ is indefinite} & \text{if } \Delta_k \text{ does not satisfy in any of the above criteria.} \end{cases}$$

2. **Eigenvalue Test:**
   - All eigenvalues of A are positive $\Rightarrow$ A is positive definite.
   - All eigenvalues of A are negative $\Rightarrow$ A is negative definite.
   - All eigenvalues are non-negative (some zero) $\Rightarrow$ positive semi-definite.
   - All eigenvalues are non-positive (some zero) $\Rightarrow$ negative semi-definite.
   - Mixed signs $\Rightarrow$ indefinite.

Problem 2 *Determine the nature of the quadratic form*

$$Q(x_1, x_2, x_3) = 2x_1^2 + 2x_2^2 + 2x_3^2 + 2x_1x_2 + 2x_2x_3 + 2x_3x_1.$$

Solution: The corresponding symmetric matrix is

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

The leading principal minors are:

$$\Delta_1 = |2| = 2 > 0,$$

$$\Delta_2 = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 4 - 1 = 3 > 0,$$

$$\Delta_3 = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix} = 2(4-1) - 1(2-1) + 1(1-2) = 6 - 1 - 1 = 4 > 0.$$

Since all leading principal minors are positive, i.e.

$$\Delta_1 > 0, \quad \Delta_2 > 0, \quad \Delta_3 > 0,$$

the matrix A is **positive definite**. Hence, the quadratic form $Q(x_1, x_2, x_3)$ is **positive definite**.

Problem 3 *Determine the nature of the quadratic form*

$$Q(x_1, x_2, x_3) = x_1^2 + 2x_2^2 - 3x_3^2 + 2x_1x_2 - 2x_2x_3.$$

Solution: The corresponding symmetric matrix is

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & -3 \end{bmatrix}.$$

Compute the leading principal minors:

$$\Delta_1 = |1| = 1 > 0,$$

$$\Delta_2 = \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} = (1)(2) - (1)(1) = 1 > 0,$$

$$\Delta_3 = \begin{vmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & -3 \end{vmatrix} = 1(2 \times (-3) - (-1) \times (-1)) - 1(1 \times (-3) - 0 \times (-1)) + 0(\cdots)$$
$$= 1(-6-1) - 1(-3) = (-7) + 3 = -4 < 0.$$

Thus,

$$\Delta_1 > 0, \quad \Delta_2 > 0, \quad \Delta_3 < 0.$$

Since the leading principal minors do not all have the same sign, the matrix A is **indefinite**. Hence, the quadratic form $Q(x_1, x_2, x_3)$ is **indefinite** (it takes both positive and negative values).

Problem 4 *Check the nature of the quadratic form by eigenvalue method*

$$Q(x_1, x_2, x_3) = 2x_1^2 + 2x_2^2 + 2x_3^2 + 2x_1x_2 + 2x_2x_3 + 2x_3x_1.$$

Solution: The corresponding symmetric matrix is

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

The characteristic equation is obtained from

$$|A - \lambda I| = 0.$$

$$\begin{vmatrix} 2-\lambda & 1 & 1 \\ 1 & 2-\lambda & 1 \\ 1 & 1 & 2-\lambda \end{vmatrix} = 0.$$

Expanding,

$$(2-\lambda)\big[(2-\lambda)^2 - 1\big] - 1\big[(1)(2-\lambda) - 1\big] + 1\big[(1) - (2-\lambda)\big] = 0.$$

Simplifying gives

$$(2-\lambda)\big[(2-\lambda)^2 - 1\big] = 0.$$

$$(2-\lambda)(1-\lambda)(3-\lambda) = 0.$$

Hence, the eigenvalues are

$$\lambda_1 = 1, \quad \lambda_2 = 2, \quad \lambda_3 = 3.$$

Since all eigenvalues are positive, the matrix A is **positive definite**. Therefore, the quadratic form $Q(x_1, x_2, x_3)$ is **positive definite**.

4. Gradient and Hessian

For the quadratic function

$$f(x) = \frac{1}{2}x^T Q x + c^T x + d,$$

the first and second derivatives are:

$$\nabla f(x) = Qx + c, \qquad H = \nabla^2 f(x) = Q.$$

5. Optimality Condition

For an unconstrained minimization problem, the necessary condition for an extremum is

$$\nabla f(x^*) = 0.$$

Hence,

$$Qx^* + c = 0 \quad \Rightarrow \quad x^* = -Q^{-1}c,$$

provided Q is nonsingular.

6. Hessian Matrix of Non-Quadratic Form

Let $f(x_1, x_2, \ldots, x_n)$ be a real-valued function (not necessarily quadratic) having continuous second-order partial derivatives. Then, the *Hessian matrix* of f is defined as the square matrix of all second-order partial derivatives of f, denoted by $H(f)$ or simply H.

$$H(f) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

If the function f has continuous second-order partial derivatives, then by Clairaut's theorem the mixed partial derivatives are equal, that is,

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}, \quad \forall i, j,$$

and hence the Hessian matrix $H(f)$ is **symmetric**.

7. Nature of the Stationary Point

For an unconstrained minimization problem, the necessary condition for an extremum is

$$\nabla f(x^*) = 0.$$

The **Hessian matrix** $H(f)$ determines the nature of the stationary point x^*.

- If $H(f)$ is **positive definite**, x^* is a **strict local (and global) minimum**.
- If $H(f)$ is **positive semidefinite**, x^* is a **local (possibly non-unique) minimum**.
- If $H(f)$ is **negative definite**, x^* is a **strict local (and global) maximum**.
- If $H(f)$ is **negative semidefinite**, x^* is a **local (possibly non-unique) maximum**.
- If $H(f)$ is **indefinite**, x^* is a **saddle point**.

The definiteness of the matrix $H(f)$ can be checked by the techniques given in Section 3.

Problem 5 *Consider*

$$f(x_1, x_2) = x_1^2 + x_1 x_2 + x_2^2 - 6x_1 - 9x_2.$$

Find out the stationary points and check the nature of that stationary points.

Answer: The stationary points are given by:

$$\nabla f(x) = \begin{bmatrix} 2x_1 + x_2 - 6 \\ x_1 + 2x_2 - 9 \end{bmatrix} = 0.$$

Solving, we get

$$x_1^* = 1, \quad x_2^* = 4.$$

The Hessian matrix is give by

$$H = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Then

Since all leading principal minors of H are positive:

$$\Delta_1 = 2 > 0, \quad \Delta_2 = 3 > 0,$$

Q is **positive definite**, hence $x^* = (1, 4)$ is a **minimum point**.

Problem 6 *Checking Definiteness of the Hessian Matrix of*

$$f(x_1, x_2) = \exp(x_1 + 1) + \exp(1 - x_2).$$

Answer:

$$f_{x_1} = \frac{\partial}{\partial x_1}\big(\exp(x_1 + 1)\big) = \exp(x_1 + 1),$$
$$f_{x_2} = \frac{\partial}{\partial x_2}\big(\exp(1 - x_2)\big) = -\exp(1 - x_2).$$

$$f_{x_1x_1} = \exp(x_1 + 1), \qquad f_{x_2x_2} = \exp(1 - x_2),$$
$$f_{x_1x_2} = f_{x_2x_1} = 0.$$

Hessian Matrix

$$H(f) = \begin{pmatrix} f_{x_1x_1} & f_{x_1x_2} \\ f_{x_2x_1} & f_{x_2x_2} \end{pmatrix} = \begin{pmatrix} \exp(x_1 + 1) & 0 \\ 0 & \exp(1 - x_2) \end{pmatrix}.$$

Definiteness of the Hessian

Since the exponential function is always positive,

$$\exp(x_1 + 1) > 0, \qquad \exp(1 - x_2) > 0 \quad \forall (x_1, x_2) \in \mathbb{R}^2.$$

The determinant of the Hessian is

$$\det(H) = \exp(x_1 + 1)\,\exp(1 - x_2) > 0.$$

Thus, the Hessian matrix is **positive definite** for all (x_1, x_2), and therefore the function f is **strictly convex**.

Convex and Concave Functions

Definition: Convex Function

A function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be *convex* on a convex set D if for every $x, y \in D$ and for all $\lambda \in [0, 1]$, we have

$$f\big(\lambda x + (1-\lambda)y\big) \leq \lambda f(x) + (1-\lambda)f(y).$$

Geometrically, this means that the graph of f lies *below* the straight line segment joining the points $(x, f(x))$ and $(y, f(y))$.

Example

Consider the function

$$f(x) = x^2, \quad x \in \mathbb{R}.$$

The second derivative is $f''(x) = 2 > 0$, hence the function is strictly convex. Graphically, the parabola opens upward.

Definition: Concave Function

A function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be *concave* on a convex set D if for every $x, y \in D$ and for all $\lambda \in [0, 1]$, we have

$$f\big(\lambda x + (1-\lambda)y\big) \geq \lambda f(x) + (1-\lambda)f(y).$$

Geometrically, the graph of f lies *above* the line segment joining $(x, f(x))$ and $(y, f(y))$.

Example

Consider the function

$$f(x) = \sqrt{x}, \quad x > 0.$$

Since

$$f''(x) = -\frac{1}{4x^{3/2}} < 0,$$

the function is strictly concave on $(0, \infty)$.

Remark on Convexity and the Hessian Matrix

For a twice differentiable function $f : \mathbb{R}^n \to \mathbb{R}$, the nature of its convexity or concavity can be determined using the Hessian matrix $H_f(x)$.

- If the Hessian matrix $H_f(x)$ is *positive semidefinite* for all x in a convex domain D, then the function f is **convex** on D.
- If the Hessian matrix $H_f(x)$ is *positive definite* for all $x \in D$, then f is **strictly convex** on D.
- If the Hessian matrix $H_f(x)$ is *negative semidefinite* for all $x \in D$, then f is **concave** on D.

- If the Hessian matrix $H_f(x)$ is *negative definite* for all $x \in D$, then f is **strictly concave** on D.

Thus, the definiteness of the Hessian provides a convenient and powerful criterion for analyzing the convexity or concavity of smooth multivariable functions.

Remark on Convexity and Maxima–Minima

The convexity properties of a function play a fundamental role in determining the nature of its stationary points:

- If a differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is **convex** on a convex set D, then any point x^* satisfying $\nabla f(x^*) = 0$ is necessarily a **global minimizer** of f on D. No other local minima exist.
- If f is **strictly convex** on D, then the global minimizer is **unique**.
- If the function f is **concave**, then any point where $\nabla f(x^*) = 0$ is a **global maximizer** on D.
- A convex function cannot have a nontrivial local maximum, and a concave function cannot have a nontrivial local minimum.

Thus, the convexity or concavity of a function allows one to establish global optimality from first-order conditions alone.

Problem 7 *Obtain all the stationary points of the following function and describe the nature of them.*

$$f(x, y) = x^3 + y^3 + 2x^2 + 4y^2 + 6.$$

Solution.

We consider the function

$$f(x, y) = x^3 + y^3 + 2x^2 + 4y^2 + 6.$$

Since, the function is not quadratic, so all the extrema is local extrema.

Stationary points

$$f_x = 3x^2 + 4x, \qquad f_y = 3y^2 + 8y.$$

Set $f_x = 0$ and $f_y = 0$.

$$3x^2 + 4x = 0 \quad \Rightarrow \quad x(3x + 4) = 0$$

Thus

$$x = 0, \qquad x = -\frac{4}{3}.$$

Similarly,

$$3y^2 + 8y = 0 \quad \Rightarrow \quad y(3y + 8) = 0$$

Thus

$$y = 0, \qquad y = -\frac{8}{3}.$$

Hence the stationary points are

$$(0,0), \qquad \left(0, -\frac{8}{3}\right), \qquad \left(-\frac{4}{3}, 0\right), \qquad \left(-\frac{4}{3}, -\frac{8}{3}\right).$$

Hessian matrix

$$f_{xx} = 6x + 4, \qquad f_{yy} = 6y + 8, \qquad f_{xy} = 0.$$

Hence

$$H = \begin{pmatrix} 6x+4 & 0 \\ 0 & 6y+8 \end{pmatrix}.$$

Since H is diagonal, the definiteness depends on the signs of $6x + 4$ and $6y + 8$.

Nature of each stationary point

(1) Point $(0, 0)$

$$\Delta_1 = 4 > 0, \qquad \Delta_2 = 32 > 0.$$

Both the principle minors are positive, so the Hessian matrix H is positive definite at $(0, 0)$

$$\Rightarrow (0,0) \text{ local minimum.}$$

(2) Point $\left(0, -\frac{8}{3}\right)$

$$\Delta_1 = 4 > 0, \qquad \Delta_2 = 4(-8) = -32 < 0.$$

So, H is indefinite.

$$\Rightarrow \left(0, -\frac{8}{3}\right) \text{ saddle point.}$$

(3) Point $\left(-\frac{4}{3}, 0\right)$

$$\Delta_1 = -4 < 0, \qquad \Delta_2 = -32 > 0.$$

So, H is indefinite.

$$\Rightarrow \text{Saddle point.}$$

(4) Point $\left(-\frac{4}{3}, -\frac{8}{3}\right)$

$$\Delta_1 = -4 < 0, \qquad \Delta_2 = 32 > 0.$$

H is negative definite.

$$\Rightarrow \text{Local maximum.}$$

Final classification

$$\begin{aligned} (0,0) &: \text{ Local minimum,} \\ \left(0, -\tfrac{8}{3}\right) &: \text{ Saddle point,} \\ \left(-\tfrac{4}{3}, 0\right) &: \text{ Saddle point,} \\ \left(-\tfrac{4}{3}, -\tfrac{8}{3}\right) &: \text{ Local maximum.} \end{aligned}$$

Problem 8 *Obtain all the stationary points of the following function and describe the nature of them.*

$$f(x_1, x_2, x_3) = x_1 + 2x_3 + x_2x_3 - x_1^2 - x_2^2 - x_3^2.$$

Solution.
Consider the function

$$f(x_1, x_2, x_3) = x_1 + 2x_3 + x_2x_3 - x_1^2 - x_2^2 - x_3^2.$$

First–order partial derivatives

$$f_{x_1} = 1 - 2x_1, \qquad f_{x_2} = x_3 - 2x_2, \qquad f_{x_3} = 2 + x_2 - 2x_3.$$

Stationary points
Set the first derivatives equal to zero:

$$\begin{gathered} 1 - 2x_1 = 0 \quad \Rightarrow \quad x_1 = \frac{1}{2}. \\ x_3 - 2x_2 = 0 \quad \Rightarrow \quad x_3 = 2x_2. \\ 2 + x_2 - 2x_3 = 0. \end{gathered}$$

Substituting $x_3 = 2x_2$ into the last equation:

$$\begin{gathered} 2 + x_2 - 4x_2 = 0 \\ 2 - 3x_2 = 0 \quad \Rightarrow \quad x_2 = \frac{2}{3}. \end{gathered}$$

Then

$$x_3 = 2x_2 = \frac{4}{3}.$$

Hence the unique stationary point is

$$\boxed{\left(\frac{1}{2}, \frac{2}{3}, \frac{4}{3}\right)}.$$

Hessian matrix

$$H = \begin{pmatrix} f_{x_1x_1} & f_{x_1x_2} & f_{x_1x_3} \\ f_{x_2x_1} & f_{x_2x_2} & f_{x_2x_3} \\ f_{x_3x_1} & f_{x_3x_2} & f_{x_3x_3} \end{pmatrix} = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 1 & -2 \end{pmatrix}.$$

Definiteness of the Hessian
Compute the leading principal minors:

$$\begin{gathered} \Delta_1 = -2 < 0, \\ \Delta_2 = \begin{vmatrix} -2 & 0 \\ 0 & -2 \end{vmatrix} = 4 > 0, \\ \Delta_3 = \begin{vmatrix} -2 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 1 & -2 \end{vmatrix} = -2(4-1) = -6 < 0. \end{gathered}$$

Thus the signs alternate:

$$(-, +, -),$$

which implies that the Hessian matrix is **negative definite**.

Nature of the stationary point

Since the Hessian is negative definite, f is strictly concave near the stationary point. Therefore,

$$\boxed{\left(\frac{1}{2}, \frac{2}{3}, \frac{4}{3}\right) \text{ is a local maximum.}}$$